

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1945

Siloam Springs 138 kV
January 26, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1945. The Interconnection Customer proposes a 200.0 MW solar photovoltaic facility interconnecting at the Siloam Springs 138 kV substation in the Virginia area. The request was received on November 29, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 433 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$32,816,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: VAN TYLE-CHOLAME 138.0 kV
Contingency: SILOAM SPRINGS-LEWISBURG 138.0 kV Line
Loading: 125.1% (Rating: 220 MVA)
DFAX: 36.3%
Pre-project loading: 92.1%
2. Facility: LEWISBURG-S JORDAN 138.0 kV
Contingency: MIDDLETOWN JCT-MOUNT PLEASANT 138.0 kV L
Loading: 118.0% (Rating: 343 MVA)
DFAX: 21.4%
Pre-project loading: 105.6%
3. Facility: VAN TYLE-RIVERTON 230 138.0 kV L
Contingency: SILOAM SPRINGS-LEWISBURG 138.0 kV
Loading: 115.2% (Rating: 344 MVA)
DFAX: 23.4%
Pre-project loading: 101.6%
4. Facility: CHAFFEE GAS RECOVERY-SOUTH JORDAN 138.0 kV
Contingency: PASCO CNTY SOLID WASTE RESOURCE RECOVERY-WHEELER 138.0 kV
Loading: 110.0% (Rating: 415 MVA)
DFAX: 10.2%
Pre-project loading: 105.1%
5. Facility: NOVEC PARADISE-PASCO CNTY SOLID WASTE RESOURCE RECOVERY 138.0 kV
Contingency: S JORDAN-CHAFFEE GAS RECOVERY 138.0 kV
Loading: 105.6% (Rating: 391 MVA)
DFAX: 6.1%
Pre-project loading: 102.5%

6. Facility: SILOAM SPRINGS-LEWISBURG 138.0 kV L
 Contingency: FT BENTON-DETOUR 138.0 kV
 Loading: 104.4% (Rating: 437 MVA)
 DFAX: 33.4%
 Pre-project loading: 89.1%
7. Facility: THOMASTON-TOA VACA 138.0 kV
 Contingency: CHAFFEE GAS RECOVERY-BLAINE 138.0 kV
 Loading: 104.3% (Rating: 343 MVA)
 DFAX: 8.3%
 Pre-project loading: 99.5%
8. Facility: CHAFFEE GAS RECOVERY-BLAINE 138.0 kV
 Contingency: W STAUNTON-WOODS DP 138.0 kV L
 Loading: 101.6% (Rating: 371 MVA)
 DFAX: 8.0%
 Pre-project loading: 97.3%
9. Facility: BLAINE-STONEWALL ENERGY CTR 138.0 kV L
 Contingency: WEST STAUNTON-WOODS DP 138.0 kV
 Loading: 100.7% (Rating: 329 MVA)
 DFAX: 8.0%
 Pre-project loading: 95.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VAN TYLE-CHOLAME 138.0 kV	Rebuild line with higher-capacity conductor	\$24,359,000	90% (\$21,809,000)
LEWISBURG-S JORDAN 138.0 kV	Rebuild line with higher-capacity conductor	\$5,089,000	71% (\$3,617,000)
VAN TYLE-RIVERTON 230 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,879,000	100% (\$2,879,000)
SOUTH JORDAN-CHAFFEE GAS RECOVERY 138.0 kV L	Reconductor line	\$7,243,000	29% (\$2,118,000)
PASCO CNTY SOLID WASTE RESOURCE RECOVERY SUBSTATION-NOVEC PARADISE 138.0 kV	Reconductor line	\$1,985,000	18% (\$359,000)
LEWISBURG-SILOAM SPRINGS SUBSTATION 138.0 kV	Reconductor line	\$2,770,000	58% (\$1,621,000)
THOMASTON-TOA VACA 138.0 kV	Reconductor line	\$741,000	8% (\$62,000)
CHAFFEE GAS RECOVERY-BLAINE 138.0 kV	Reconductor line	\$728,000	32% (\$236,000)
STONEWALL ENERGY CTR-BLAINE 138.0 kV	Reconductor line	\$356,000	32% (\$115,000)
	Total allocated to project		\$32,816,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.